st_04_principles

1. Abstractions, Examples

Please, visit https://www.studentenwerk-leipzig.de/mensen-cafeterien/speiseplan. Read the various offers for meals carefully. Develop categorization schemes for each meal offer in its collapsed state. Provide a structural description for each categorization scheme you identified. Please verify, that each meal offer is covered by one of your categorization schemes. If you describe your categories using classes do not use inheritance!

2. Abstraction Levels, OMG Four Layers Standard Modeling Stack

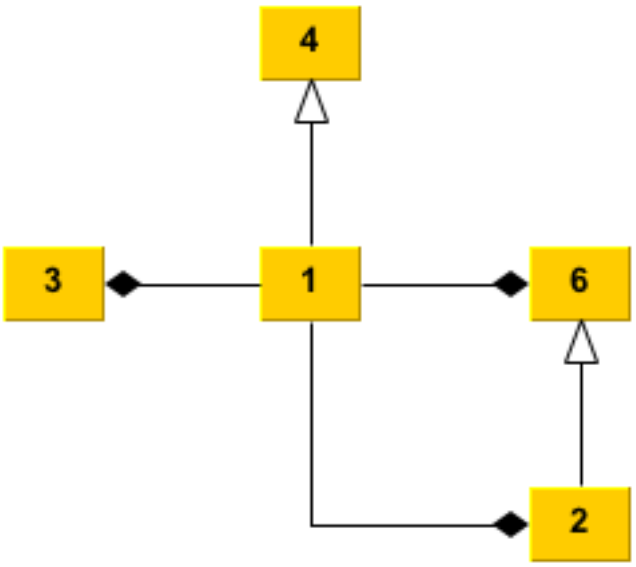
1. Hopefully, you identified more than one category in the previous exercise. Now look at the vocabulary used for the descriptions. Try to categorize the vocabulary, name the categories, and explicitly map your vocabulary to these categories.
2. Now have a look at the slide describing the OMG four layers standard modeling stack. Which is the appropriate level for each of the following:
    a. the meal offers from https://www.studentenwerk-leipzig.de/mensen-cafeterien/speiseplan,
    b. your categorization schemes for the meal offers,
    c. and the categories for vocabulary identified in the previous subtask?

3. Structuring

The first paragraph of the Wikipedia article “Blockchain” states verbatim “*Eine Blockchain … ist eine kontinuierlich erweiterbare Liste von Datensätzen, genannt „Blöcke“, welche mittels kryptographischer Verfahren miteinander verkettet sind. … Jeder Block enthält dabei typischerweise einen kryptographisch sicheren Hash (Streuwert) des vorhergehenden Blocks, … einen Zeitstempel und Transaktionsdaten.*“ (https://de.wikipedia.org/wiki/Blockchain, accessed 10/23/2018). Please read this paragraph carefully, then tell which parts are connected by which relation. Refer to the preceding text exclusively! Draw a simple diagram.

4. Hierarchization

Look at the following UML class diagram.



1. Which hierarchies does it contain?
2. Now assume, that each class has a constructor which prints the class name. If a class A inherits from another class B, then the constructor of B is called before the constructor of A is executed. If a class C aggregates a class D, then the constructor of D is called before the constructor of C is executed. Furthermore, inheritance has a higher priority than aggregation with respect to constructor invocation. Given all that, write down the resulting cipher sequence when creating one exemplar of each class followed a newline character.

5. Hierarchization

Given are the following sentences which are produced by creating exemplars of different classes, each creation followed by a newline character, that is, the same mechanism as outlined in the preceding task:

A
AOP
AOPM
AOPMGIY
GI

Hint: It is possible that a class name has more than one character!

6. Modularization

Please, discuss whether a function can be appropriately classified as module! Can you conceive modules on a smaller scale than a function?

7. Modularization

The following list shows small software entities and the connections among them:

1. A: A, D, X
2. B: B, J, L, N, O
3. C: A, C, D, N, X
4. D: D, J, K
5. J: D, M
6. K: D, M
7. L: B, L, N, O
8. M: J, K, N
9. N: B, C, L, O
10. O: B, L, N, O
11. X: A, C, D

“J: D, M” means, J and D are connected and J and M are connected. There is no information available, whether these connections are directed or not. Thus, you should not make any assumption concerning direction.

1. First, you should think about a plausibility check for the list above and apply it. Is the information in the list consistent? If not, try to repair it. Doing so, you may add items, but not remove them.
2. Next, think about another more explicit and more formal representation for the information contained in the list. Maybe you consider a table or a matrix.
3. Enter the information of the list into the new representation. Then analyze the entities and their connections. Afterwards you should try to transform the content of the new representation in a consistent way, in order to recognize groups or clusters. Now make a proposal how to modularize them. How many modules do result? Which entities are placed into which module? Are there connections among the modules you identified? If so, which? Motivate your decisions by applying the criteria cohesion and coupling for good modularization.

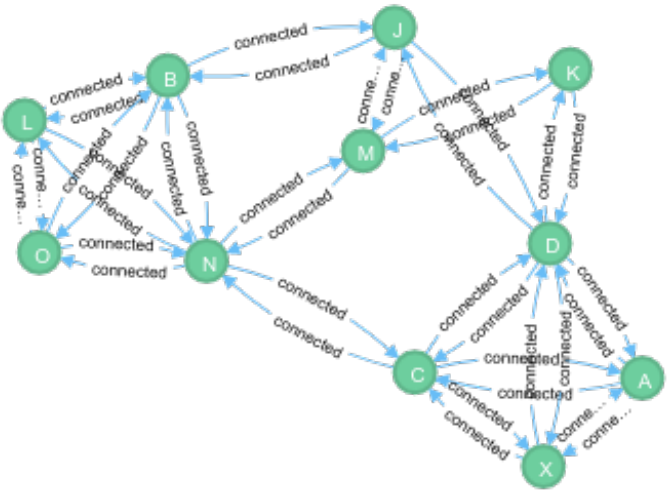
8. Rigor and Formality, Generalization

Revisit the task of modularization of entities and their connection and have a look at your solution and the solution presented by your exercise instructors. What is your guess, how high the effort is for finding a formal solution that can be applied to the complete class of similar problems?

Nico Willert Stark and Felix Nirsberger, both former exercise instructors, thought about approximations to formal solutions using different paradigms. Nico came up with some kind of similarity measure and swapping of rows and columns of a matrix until 1’s were agglutinated.

	d	c	a	x	j	k	n	b	l	o	m
d	1	1	1	1	1	1	0	0	0	0	0
c	1	1	1	1	0	0	1	0	0	0	0
a	1	1	1	1	0	0	0	0	0	0	0
x	1	1	1	0	0	0	0	0	0	0	0
j	1	0	0	0	0	0	0	1	0	0	1
k	1	0	0	0	0	0	0	0	0	0	1
n	0	1	0	0	0	0	0	1	1	1	1
b	0	0	0	0	1	0	1	1	1	1	0
l	0	0	0	0	0	0	1	1	1	1	0
o	0	0	0	0	0	0	1	1	1	1	0
m	0	0	0	0	1	1	1	0	0	0	0

Felix fed the information about connection into a graph database and displayed the result as a graph.



Is it in this case appropriate to devise such a formal solution to this problem? What is your opinion?

9. Information hiding

Have a closer look at classes in C++ and Java and access specifiers. Discuss which variant of information hiding is applied in both programming languages.

10. Verbalization

Check some source code of your choice. Look for good and bad examples of verbalization in comments, documentation comments, user defined identifiers, and reserved words of the programming language. You should present in total eight examples covering each combination described before.

11. Other Principles

Do you know what an obfuscator is? If not, inform yourself about its purpose and how it works. Find at least one example for an obfuscator. Which principle is affected most by applying an obfuscator?

12. All principles

Checkout risk_assessment.pl from the repository. This program was kindly developed by Andreas Barton, head of faculty service center. Please, assess this program according to the principles mentioned in the slides. For each principle you should state, whether it is applicable or not and why. If it is applicable, you should write down your result of assessing the program according to this principle.

13. All principles

You are a member of a team that develops a large application using a programming language with strong static typing. The team feels, that development is not as smooth as possible. Almost each team member works on all parts of the software. Furthermore, the parameter lists of functions and methods need to be frequently adjusted to externally triggered changes. Thus, an external consultant had a closer look to your project. He argues, that locality is bad and that the frequent changes, which cannot be foreseen exactly, are not properly anticipated. As a remedy he proposes a fundamental design change. Functions and methods should be rewritten, so that they take only one parameter. This parameter should encapsulate all the previously individually passed parameters. The return value of a function or method can be a compound type too, if necessary. This way, the number of a parameters is kept fixed, i.e., one. If changes are necessary they are applied in one place only, the single parameter passed.
Now the proposal is discussed by the team. What is your opinion and how do you argue?